

Scalability & Future Plan

Date	15 July 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Supports only local Flask deployment	Cannot be accessed by users over the Internet	High
2	SQLite database is suitable only for small datasets	Limited scalability for large numbers of users	Medium
3	No user authentication or login system	Anyone can access the application	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a single local user	Deploy on Render, Railway, or Azure to support multiple users
2	Data Storage	SQLite database	Upgrade to MySQL or PostgreSQL
3	Performance	Random Forest model runs locally	Optimize model and use cloud-based deployment
4	Security	No authentication	Add user login, password encryption, and HTTPS

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Deploy the Flask application on a cloud platform (Render/Azure/Railway)	1 Month	Enables online access and supports multiple users
Phase 2	Implement user authentication, secure login, and migrate from SQLite to MySQL	2-3 Months	Improves security, reliability, and database scalability
Phase 3	Develop a mobile application, REST API integration, and enhance the ML model with additional financial parameters	4-6 Months	Improves accessibility, prediction accuracy, and supports future expansion